



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0510 – COMPUTER VISION

Name: Premarubha N

EXPERIMENT NO: 19

Reg No: 192525020

19. Perform Edge detection using Sobel Matrix along XY axis

AIM:

Edge detection using Sobel Matrix along XY axis

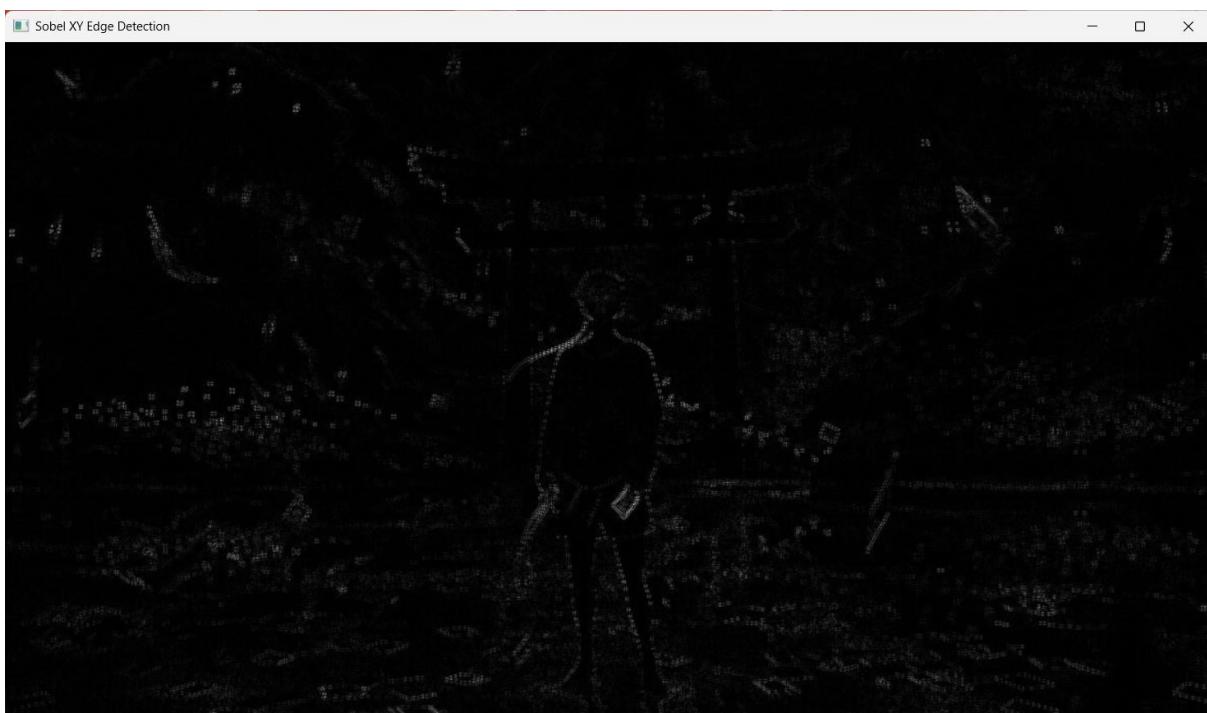
PROGRAM:

```
import cv2
img = cv2.imread(r"C:\Users\premarubha\Desktop\cv model image.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
sobelxy = cv2.Sobel(gray, cv2.CV_64F, 1, 1, ksize=3)
sobelxy = cv2.convertScaleAbs(sobelxy)
cv2.imshow("Original Image", img)
cv2.imshow("Sobel XY Edge Detection", sobelxy)
cv2.imwrite(r"C:\Users\premarubha\Desktop\sobel_xy_image.jpg", sobelxy)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.